

Human-Object Interaction Detection: Recognizing Instantaneous Interaction Frames in Ball-Throwing Footage

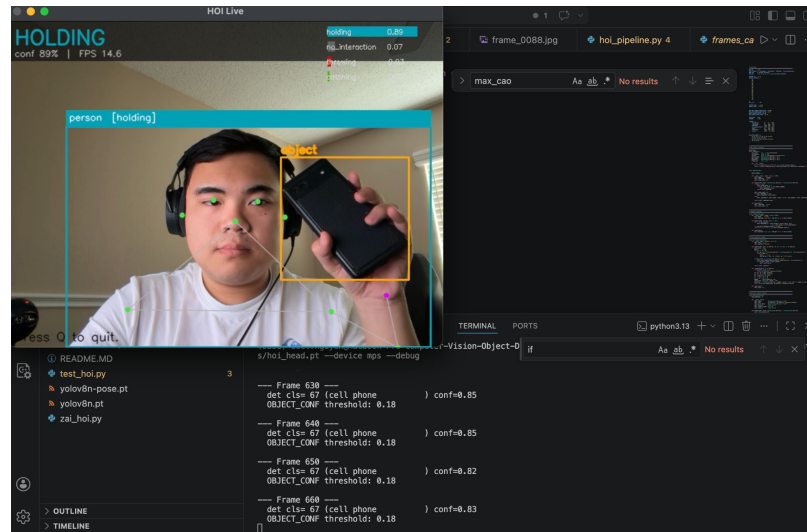
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Overview

Our primary goal was to determine the exact moment that an object is thrown by a human. To accomplish this, our methodology involves few-shot fine-tuning with both a one-pass and two-pass approach as well as prompt-engineering through a VLM.

One Pass Approach

- Built one pass Human Object Interaction (HOI) detector model on a YOLOv8 model
- Attempts to classify if a person is throwing, catching, holding, or no interaction with a object using a singular webcam
- Model:
 - Runs two YOLO models in parallel on each frame to detect a person and a object and calculates using a crop of the object and person alongside pose keypoints and spatial features
 - Uses inputs of person and object information to classify them into either throwing, catching, holding, or no interaction
 - Outputs classification and confidence level of each classification including the object classification based on object classes and the person interaction with the object



Two Pass Approach

Built a two-pass HOI (Human-object-interaction) model that specializes in the classification of throwing actions.

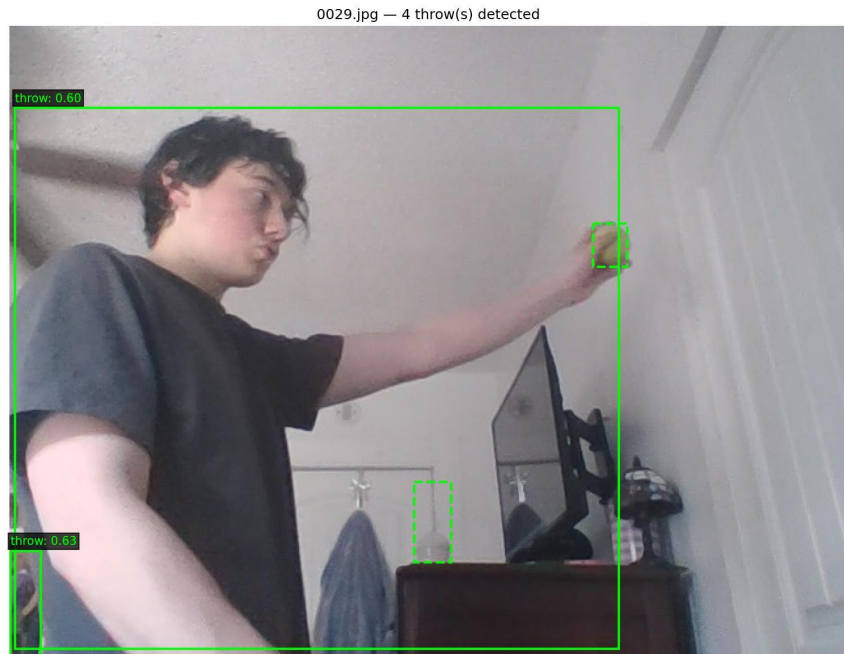
Dataset: HICO-DET, used weight sampling and heavy penalty for false negatives (since there were only 336 throwing samples)

Pass 1:

- Backbone of Faster-RCNN
- Detects persons and objects in frame first
- Enumerates all (human, object) candidate pairs

Pass 2:

- Two-stream head for interaction classification
 - Appearance stream, using RoI-aligned ResNet50 features
 - RoI: Two-channel 64*64 binary mask that encodes relative position and scale of human and object - used to capture the geometrice signature of a throwing action



Vision Language Model Approach

Implemented with Qwen2.5-VL-3B-Instruct

Prompted to describe the action in frame with in terms of [subject, action, object] with freedom outside of that

Speed- ~.25fps, too slow for reasonable inference speeds

Accurate frame analysis

